

DAY-5

Jupyter day-5 Last Checkpoint: yesterday

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Visualization

Matplotlib

```
[4]: !pip install Matplotlib

Requirement already satisfied: Matplotlib in c:\users\lenovo\anaconda3\lib\site-packages (3.9.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (1.4.4)
Requirement already satisfied: numpy>=1.23 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (1.26.4)
Requirement already satisfied: packaging>=20.0 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (24.1)
Requirement already satisfied: pillow>=8 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\lenovo\anaconda3\lib\site-packages (from Matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-packages (from python-dateutil>=2.7->Matplotlib) (1.16.0)
```

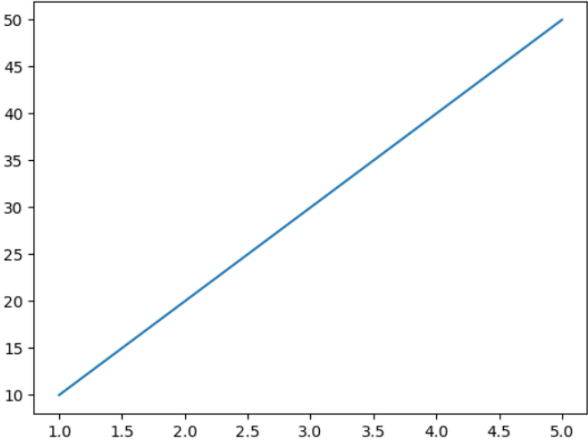
```
[5]: import matplotlib.pyplot as plt
```

```
[6]: x=[1,2,3,4,5]
     y=[10,20,30,40,50]
     plt.plot(x,y)
     plt.show()
```

Jupyter day-5 Last Checkpoint: yesterday

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Python [conda env:bas]

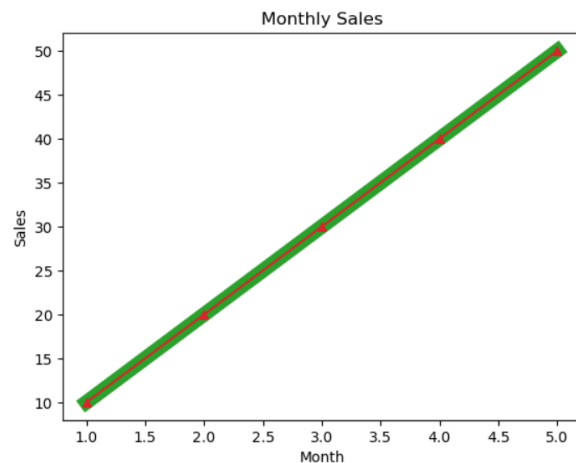


```
[26]: #Figure and Axes
      #Figure - the entire chart window
      #Axes- the sre where the graph is drawn

      import matplotlib.pyplot as plt
      fig,ax=plt.subplots()
      ax.plot([1,2,3], [10,20,30])
      #Adding a title
      plt.title("Monthly Sales")
      #adding x and y Label
      plt.xlabel("Month")
```

```
#adding markers
# 'o' - circle
# 's' - square
# '^' - triangle
# '*' - star
plt.plot(x,y,marker='^')
```

[26]: [<matplotlib.lines.Line2D at 0x17036e32e70>]



[33]: !pip install seaborn

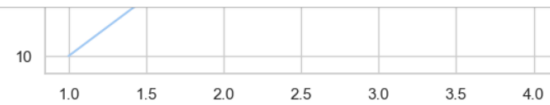
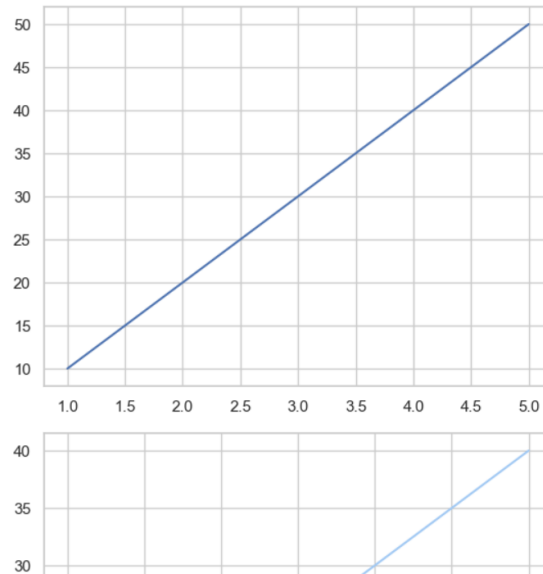
```
Requirement already satisfied: seaborn in c:\users\lenovo\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\lenovo\anaconda3\lib\site-packages (from seaborn) (1.26.4)
Requirement already satisfied: pandas>=1.2 in c:\users\lenovo\anaconda3\lib\site-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\lenovo\anaconda3\lib\site-packages (from seaborn) (3.9.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.1)
Requirement already satisfied: pillow>=8 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post 0)
Requirement already satisfied: pytz>=2020.1 in c:\users\lenovo\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\lenovo\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2023.3)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.16.0)
```

[44]:

```
import seaborn as sns
x=[1,2,3,4,5]
y=[10,20,30,40,50]
sns.set_theme()

#available styles - dark , white , ticks, darkgrid , whitegrid
sns.set_style("whitegrid")
sns.lineplot(x=x,y=y)
plt.show()
sns.set_palette("pastel") #Light color than before      #put bright - dark shade
sns.lineplot(
    x=[1,2,3,4],
    y=[10,20,30,40]
```

```
)
plt.show()
```



```
[45]: print(sns.get_dataset_names()) #to display built-in dataset of seaborn
```

```
['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes', 'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri', 'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets', 'seaiice', 'taxi', 'tips', 'titanic']
```

```
[48]: #Load dataset form titanic
df=sns.load_dataset("titanic")
print(df.head())
```

```
survived  pclass    sex  age  sibsp  parch    fare  embarked  class \
0         0         3  male  22.0     1     0   7.2500         S  Third
1         1         1  female 38.0     1     0  71.2833         C  First
2         1         3  female 26.0     0     0   7.9250         S  Third
3         1         1  female 35.0     1     0  53.1000         S  First
4         0         3  male  35.0     0     0   8.0500         S  Third

who  adult_male  deck  embark_town  alive  alone
0  man         True  NaN  Southampton  no    False
1  woman        False   C  Cherbourg  yes    False
2  woman        False  NaN  Southampton  yes    True
3  woman        False   C  Southampton  yes    False
4  man         True  NaN  Southampton  no    True
```

```
[50]: months=['Jan','Feb','Mar','Apr']
sales=[100,200,300,500]
plt.plot(months,sales,marker='o')
plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
```

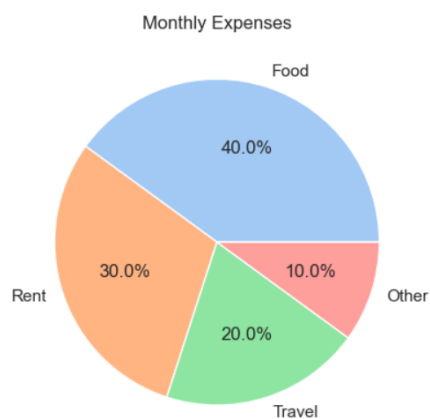
```
plt.plot(months,sales,marker='o')
plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
```

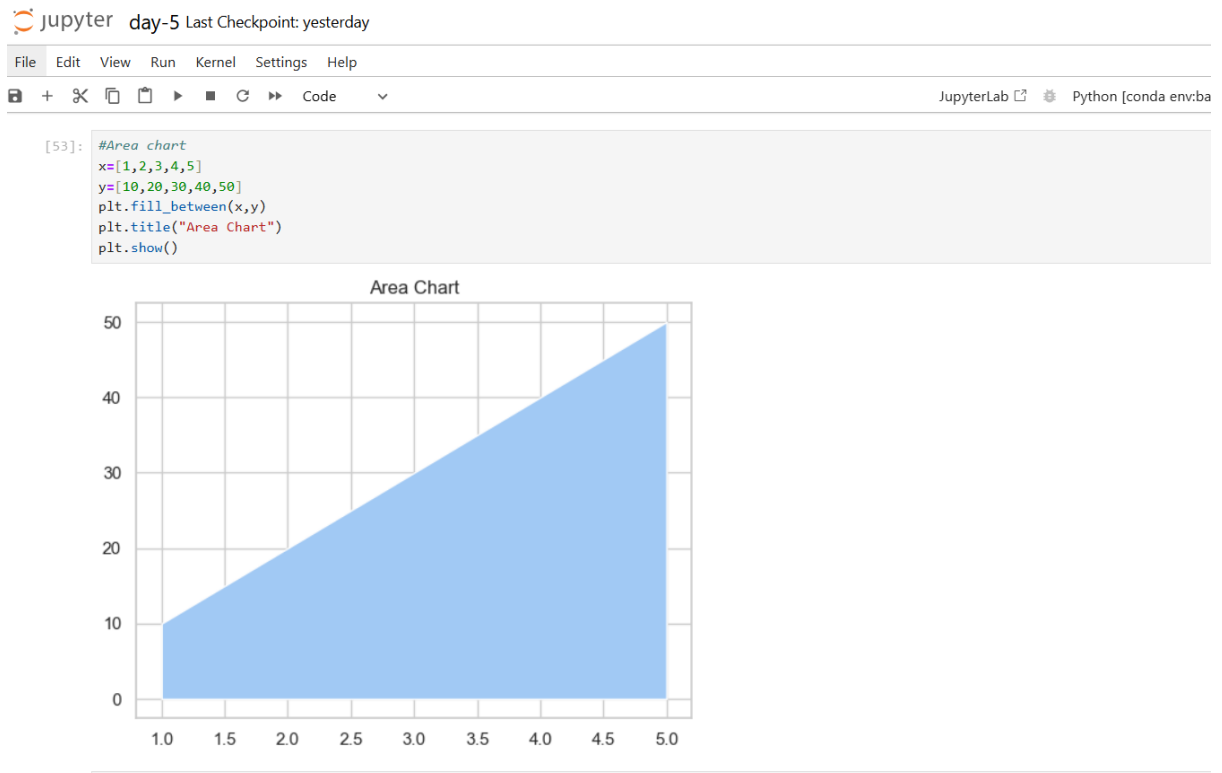
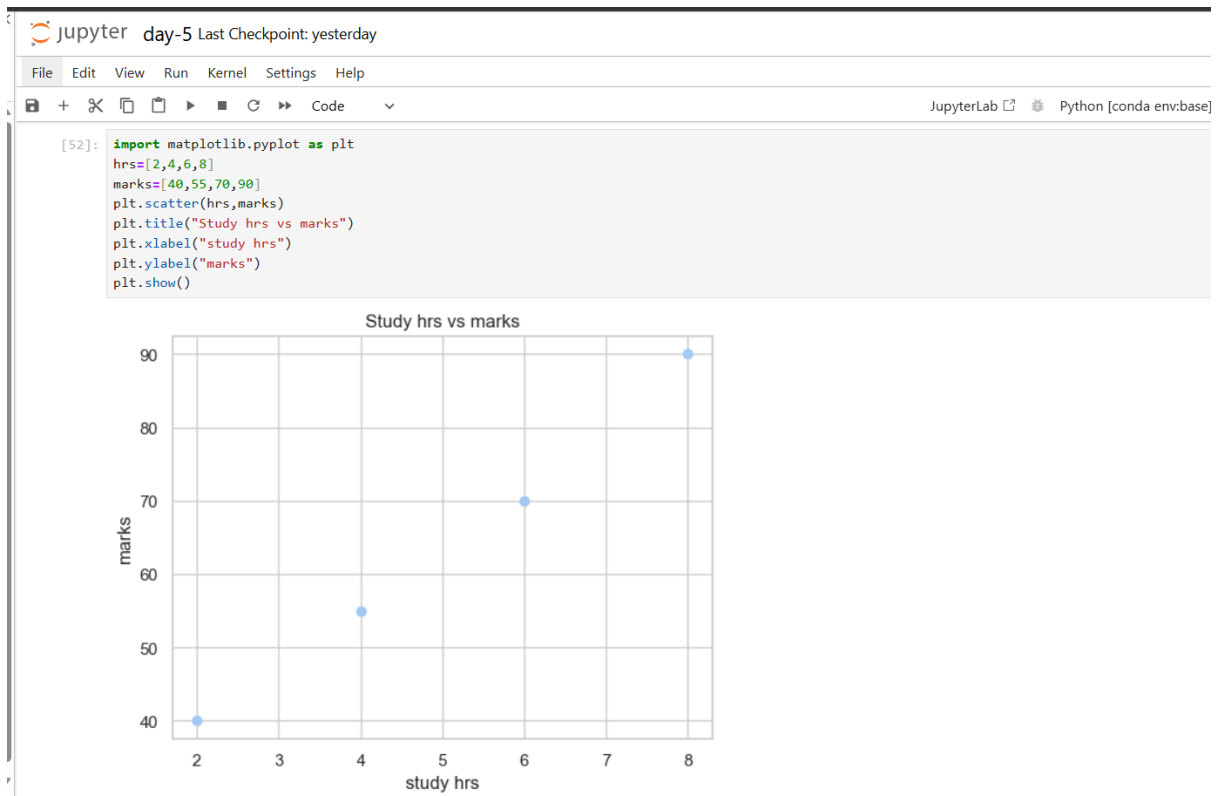
[50]: Text(0, 0.5, 'Sales')



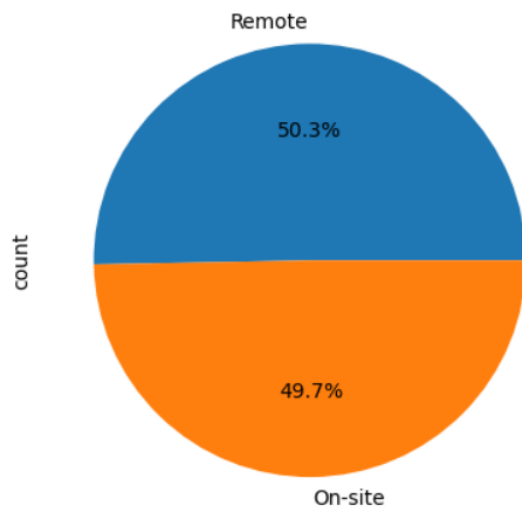
[51]: `import matplotlib.pyplot as plt`
`labels=['Food','Rent','Travel','Other']`

```
[51]: import matplotlib.pyplot as plt
labels=['Food','Rent','Travel','Other']
values=[40,30,20,10]
plt.pie(
    values,
    labels=labels,
    autopct='%1.1f%%'
)
plt.title("Monthly Expenses")
plt.show()
```

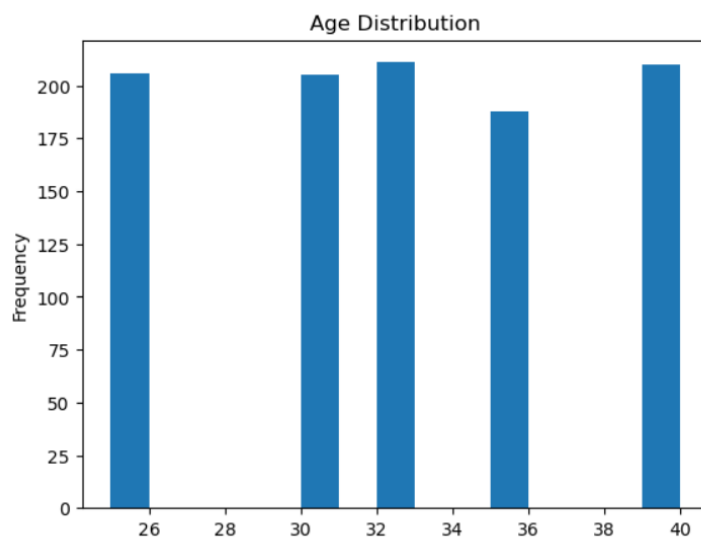




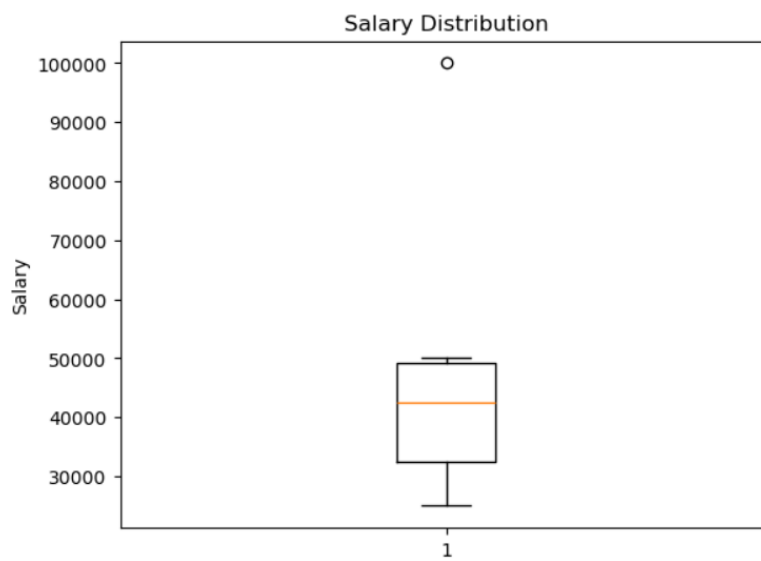
```
[12]: df['Remote_Work'].value_counts().plot(kind='pie',
      labels=['Remote', 'On-site'], autopct='%1.1f%%')
plt.show()
```



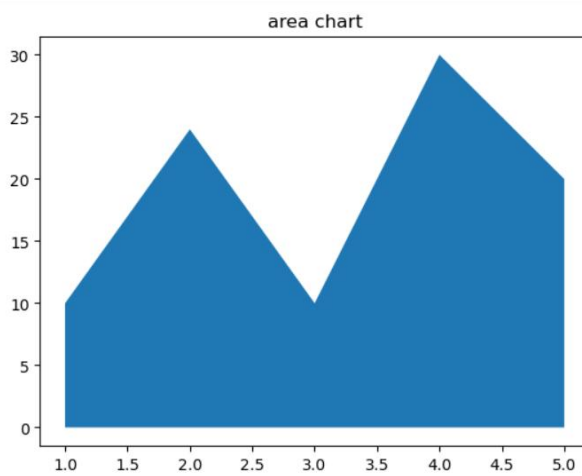
```
[13]: df['Age'].plot(kind='hist', bins=15, title='Age Distribution')
plt.show()
```



```
[14]: import matplotlib.pyplot as plt
salary=[25000,40000,35000,25000,45000,49000,50000,100000]
plt.boxplot(salary)
plt.title("Salary Distribution")
plt.ylabel("Salary")
plt.show()
```

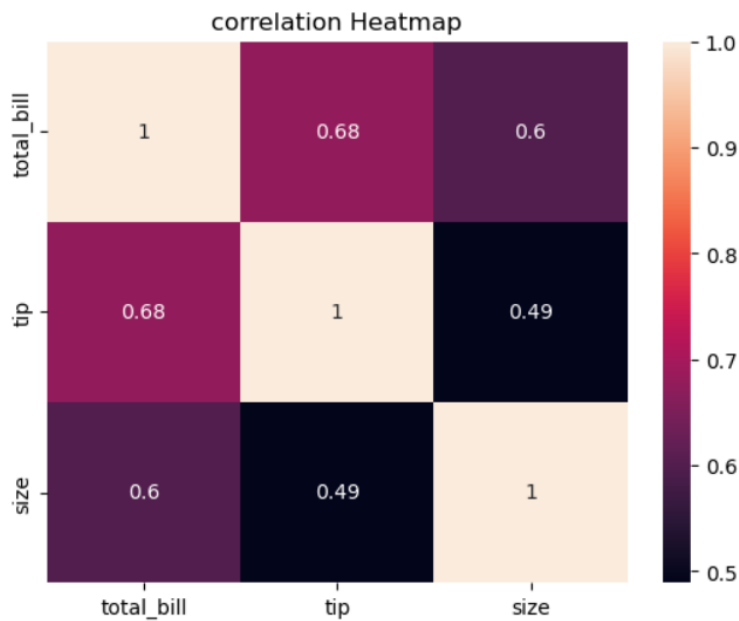


```
[20]: import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,24,10,30,20]
plt.fill_between(x,y)
plt.title("area chart")
plt.show()
```



```
[22]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
tips=sns.load_dataset("tips")
corr=tips.corr(numeric_only=True)
sns.heatmap(corr,annot=True)
plt.title("correlation Heatmap")
plt.show()
```



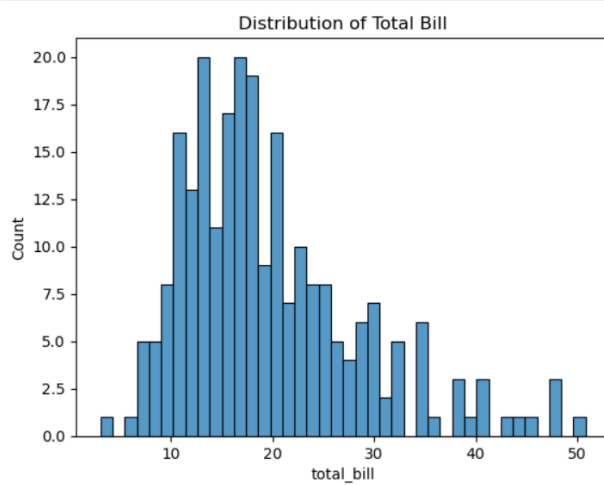
JupyterLab practice Last Checkpoint: 20 hours ago

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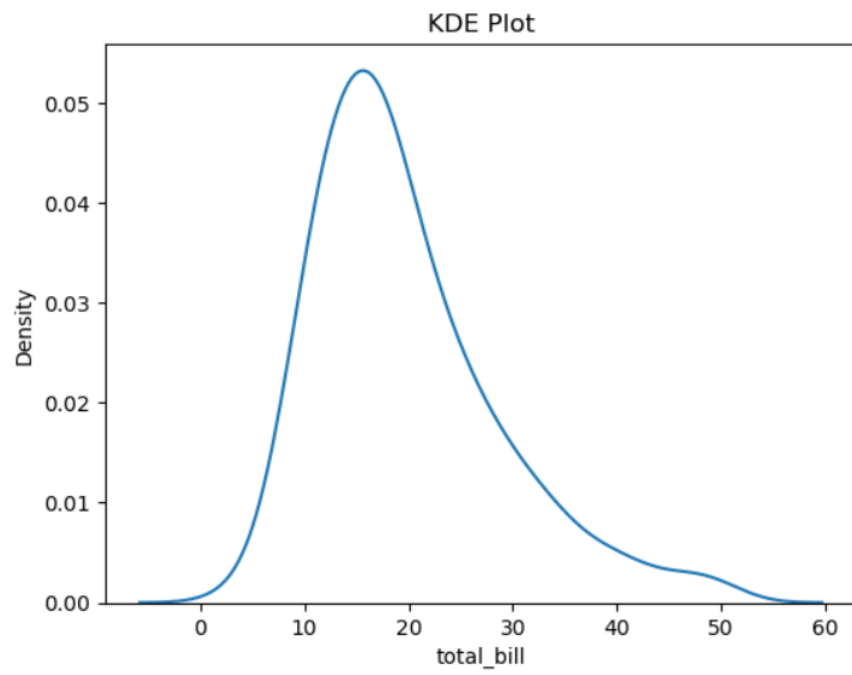
+

JupyterLab Python [conda env:base] *

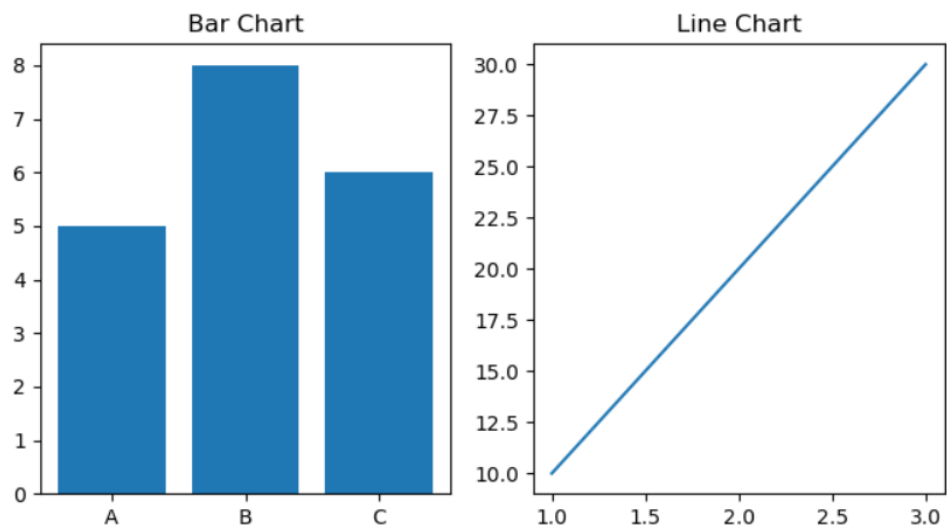
```
[23]: import seaborn as sns
sns.histplot(
    data=tips,
    x="total_bill",
    bins=40
)
plt.title("Distribution of Total Bill")
plt.show()
```




```
[24]: sns.kdeplot(  
      data=tips,  
      x="total_bill",  
      )  
plt.title("KDE Plot")  
plt.show()
```



```
plt.show()
```



```
26]: import matplotlib.pyplot as plt

plt.figure(figsize=(10,8))

plt.subplot(2,2,1)

plt.plot([1,2,3], [10,20,30])
```

